

Topic

Linear independence

⇒ Let $S = \{v_1, v_2, v_3, \dots, v_k\}$ be a set of vectors in a vector space. Then the vector equation

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$$

has at least one solution namely

$$c_1 = 0, c_2 = 0, \dots, c_k = 0$$

⇒ if this is the only solution (trivial) then S is called a linearly independent set.

⇒ if there are other solutions (non-trivial) then S is called a linearly dependent set.

Theorem:-

Let S be a set of vectors with two or more vectors. S is linearly dependent if and only if at least one of the vectors in S is expressible as a linear

Combination of the other
vectors in C .

Theorems-

o A finite set of vectors that contains the zero vector is linearly dependent.

o A set with exactly two vectors is linearly independent iff neither vector is a scalar multiple of the other.

Theorem: Linear dependent Test

$\Rightarrow$ Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in $\mathbb{R}^n$.

if $k > n$, then S is linearly dependent.

Linearly independent test by Wronskian

$\Rightarrow$ Let linearly independent of set of functions $F(-\infty, \infty)$.

Let $f_1 = f_1(x), f_2 = f_2(x), \dots, f_n = f_n(x)$

be $(n-1)$ times differentiable functions in interval $(-\infty, \infty)$, the set of the functions denoted by $C^{(n-1)}(-\infty, \infty)$.

Then the determinant of

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$$

is called the Wronskian of

$f_1, f_2, \dots, f_n$.

$\Rightarrow$ if $W(x) \neq 0$ then functions form linearly independent set of vectors.

$\Rightarrow$ if $W(x) = 0$ then L.D.

Q#: Show that the polynomials form a linearly independent set of vectors in P_n .

$\Rightarrow$ Let $P_0=1, P_1=x, P_2=x^2, \dots, P_n=x^n$ and assume

$$\therefore a_0 P_0 + a_1 P_1 + \dots + a_n P_n = 0 \quad (1)$$

are

$$a_0 \neq a_1 = a_2 = \dots = a_n = 0$$

$\Rightarrow$ But (1) is equivalent to the Statement

$$a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n = 0$$

for all x in $(-\infty, +\infty)$

$\Rightarrow$ This will be true iff each coefficient in (2) is zero.

From algebra that a non-zero polynomial of degree n has at most n distinct roots.

$\Rightarrow$ This implies that $a_0 = a_1 = a_2 = \dots = a_n = 0$ for otherwise the left side of the equation would be a non-zero polynomial with infinitely many roots. Thus, (2) has only the trivial solution.

Check whether the following vectors are linearly independent or dependent.
 $(4, 1, -2), (-4, 10, 2), (4, 0, 1)$
Let

$$V_1 = (4, 1, -2)$$

$$V_2 = (-4, 10, 2)$$

$$V_3 = (4, 0, 1)$$

$$V_1 k_1 + V_2 k_2 + V_3 k_3 = 0$$

$$(4, 1, -2)k_1 + (-4, 10, 2)k_2 + (4, 0, 1)k_3 = 0$$

$$\therefore 4k_1 - 4k_2 + 4k_3 = 0$$

$$\therefore k_1 + 10k_2 + 0k_3 = 0$$

$$\therefore -4k_1 + 2k_2 + k_3 = 0$$

$$\left[\begin{array}{ccc|c} 4 & -4 & 4 & 0 \\ 1 & 10 & 0 & 0 \\ -4 & 2 & 1 & 0 \end{array} \right]$$

$$R_1 \leftrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 4 & -4 & 4 & 0 \\ -4 & 2 & 1 & 0 \end{array} \right]$$

$$\frac{1}{4} R_2 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ -2 & 2 & 1 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ -3 & 3 & 0 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 3R_2 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 3 & 0 \end{array} \right]$$

$$\frac{1}{3} R_3 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 0 & -11 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_3 \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 0 & -11 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{R_2} \begin{bmatrix} 1 & 10 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 10R_2 \begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$\rho(A) = 3$$

$$K_1 = 0$$

$$K_2 = 0$$

$$K_3 = 0$$

$\therefore$ vectors has trivial solution.
So, the given vectors are linearly independent.

Q# Determine whether the Vectors

$$V_1 = (1, 2, 2, -1)$$

$$V_2 = (4, 9, 9, -4)$$

$$V_3 = (5, 8, 9, -5)$$

in R^n are linearly independent or dependent vectors?

$$k_1 v_1 + k_2 v_2 + k_3 v_3 = 0$$

$$k_1(1, 2, 2, -1) + k_2(4, 9, 9, -4) +$$

$$k_3(5, 8, 9, -5) = 0$$

$$\therefore k_1 + 4k_2 + 5k_3 = 0$$

$$\therefore 2k_1 + 9k_2 + 8k_3 = 0$$

$$\therefore 2k_1 + 9k_2 + 9k_3 = 0$$

$$\therefore -k_1 - 4k_2 - 5k_3 = 0$$

Coefficient matrix

$$\begin{bmatrix} 1 & 4 & 5 & : & 0 \\ 2 & 9 & 8 & : & 0 \\ 2 & 9 & 9 & : & 0 \\ -1 & -4 & -5 & : & 0 \end{bmatrix}$$

$$\begin{array}{l} R_3 \rightarrow R_3 + R_1 \\ R_3 \rightarrow R_3 - R_2 \end{array} \begin{bmatrix} 1 & 4 & 5 & : & 0 \\ 2 & 9 & 8 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \begin{bmatrix} 1 & 4 & 5 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_3 \quad \left[\begin{array}{ccc|c} 1 & 4 & 5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 4R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 5R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$K_1 = 0$$

$$K_2 = 0$$

$$K_3 = 0$$

$\therefore$ The system has trivial solution. So the given vectors are linearly independent vectors.

Determine whether the polynomials

$$P_1 = 1 - x$$

$$P_2 = 5 + 3x - 2x^2$$

$$P_3 = 1 + 3x - x^2$$

are linearly dependent or linearly independent vectors in P_2 .

$$k_1 P_1 + k_2 P_2 + k_3 P_3 = 0$$

$$\begin{aligned} &= k_1(1-x) + k_2(5+3x-2x^2) + k_3(1+3x-x^2) \\ &= (k_1 + 5k_2 + k_3) + (-k_1 + 3k_2 + 3k_3)x + \\ &\quad (-2k_2 - k_3)x^2 = 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ -1 & 3 & 3 & 0 \\ 0 & -2 & -1 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 + R_1 \left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 8 & 4 & 0 \\ 0 & -2 & -1 & 0 \end{array} \right]$$

$$R_1 \rightarrow \frac{1}{4} R_2 \left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & -2 & -1 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 5 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2 \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\boxed{k_3 = t}$$

$$2k_2 + k_3 = 0$$

$$k_1 + 3k_2 = 0$$

$$2k_2 = -t$$

$$\boxed{k_2 = -\frac{1}{2}t}$$

$$k_1 = -3 \left(-\frac{1}{2}t \right)$$

$$\boxed{k_1 = \frac{3}{2}t}$$

The system has non-trivial solution so the given polynomials are linearly dependent.

Method 2:

$$|A| = \begin{vmatrix} 1 & 3 & 3 \\ -2 & -1 & 1 \end{vmatrix} + t \begin{vmatrix} 5 & 1 \\ -2 & -1 \end{vmatrix} + 0$$

$$= (-3 + 6) + 1(-5 + 2)$$

$$= 3 - 3$$

$$= 0 \quad \therefore \text{coefficient matrix has determinant } 0$$

Determine whether the vectors in $\mathbb{R}^4$ are linearly independent or linearly dependent.
 $(1, 3, -1, -4)$, $(3, 8, -5, 7)$
 $(2, 9, 4, 23)$ (PP-2021)

Let

$$V_1 = (1, 3, -1, -4)$$

$$V_2 = (3, 8, -5, 7)$$

$$V_3 = (2, 9, 4, 23)$$

$$V_1 k_1 + V_2 k_2 + V_3 k_3 = 0$$

$$(1, 3, -1, -4)k_1 + (3, 8, -5, 7)k_2 + (2, 9, 4, 23)k_3 = 0$$

$$\begin{bmatrix} 1 \\ 3 \\ -1 \\ -4 \end{bmatrix} k_1 + \begin{bmatrix} 3 \\ 8 \\ -5 \\ 7 \end{bmatrix} k_2 + \begin{bmatrix} 2 \\ 9 \\ 4 \\ 23 \end{bmatrix} k_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 2 & : & 0 \\ 3 & 8 & 9 & : & 0 \\ -1 & -5 & 4 & : & 0 \\ -4 & 7 & 23 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1 \quad \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 3 & 8 & 9 & 0 \\ 0 & -2 & 6 & 0 \\ -4 & 7 & 23 & 0 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_4 \rightarrow R_4 + 4R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & -2 & 6 & 0 \\ 0 & 19 & 31 & 0 \end{array} \right]$$

$$\frac{1}{2} R_3 \quad \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 19 & 31 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \quad \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 19 & 31 & 0 \end{array} \right]$$

$$R_3 \leftrightarrow R_4 \quad \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 19 & 31 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 19R_2$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 88 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\frac{1}{88} R_3 \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 3R_3 \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$(-1)R_2 \rightarrow R_2 \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 3R_2 \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_1$$

$$\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$k_1 = 0$$

$$k_2 = 0$$

$$k_3 = 0$$

$\therefore$ The system has trivial solution so the given vectors are linearly independent.

Show that vectors

$$u = 2x^3 - 3x^2 - 2x + 3$$

$$v = x^3 - 4x^2 - 3x + 4$$

$$w = x^3 + 5x^2 + x + 3$$

are linearly independent (PP-2019).

Let

$$u|k_1 + v|k_2 + w|k_3 = 0$$

$$\Rightarrow (2x^3 - 3x^2 - 2x + 3)k_1 + (x^3 - 4x^2 - 3x + 4)k_2 + (x^3 + 5x^2 + x + 3)k_3 = 0$$

$$\Rightarrow (3k_1 + 4k_2 + 3k_3) + (-2k_1 - 3k_2 + k_3)x + (-3k_1 - 4k_2 + 5k_3)x^2 + (2k_1 + k_2 + k_3)x^3 = 0$$

$$3k_1 + 4k_2 + 3k_3 = 0$$

$$-2k_1 - 3k_2 + k_3 = 0$$

$$-3k_1 - 4k_2 + 5k_3 = 0$$

$$2k_1 + k_2 + k_3 = 0$$

Coefficient matrix:-

$$\begin{bmatrix} 3 & 4 & 3 & : & 0 \\ -2 & -3 & 1 & : & 0 \\ -3 & -4 & 5 & : & 0 \\ 2 & 1 & 1 & : & 0 \end{bmatrix}$$

$$\begin{array}{l} R_4 \rightarrow R_4 + R_2 \\ R_3 \rightarrow R_3 + R_1 \end{array} \begin{bmatrix} 3 & 4 & 3 & : & 0 \\ -2 & -3 & 1 & : & 0 \\ 0 & 0 & 8 & : & 0 \\ 0 & -2 & 2 & : & 0 \end{bmatrix}$$

$$\begin{array}{l} \sim \frac{1}{2} R_4 \rightarrow R_4 \\ \sim \frac{1}{8} R_3 \rightarrow R_3 \end{array} \begin{bmatrix} 3 & 4 & 3 & : & 0 \\ -2 & -3 & 1 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & -1 & 1 & : & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_3 \begin{bmatrix} 3 & 4 & 3 & : & 0 \\ -2 & -3 & 1 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & -1 & 0 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$\begin{bmatrix} 3 & 4 & 3 & : & 0 \\ 1 & 1 & 4 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & -1 & 0 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_4$$

$$\begin{bmatrix} 3 & 4 & 3 & : & 0 \\ 1 & 0 & 4 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & -1 & 0 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_3$$

$$\begin{bmatrix} 3 & 4 & 3 & : & 0 \\ 1 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & -1 & 0 & : & 0 \end{bmatrix}$$

$$(-1)R_4 \rightarrow R_4$$

$$\begin{bmatrix} 3 & 4 & 3 & : & 0 \\ 1 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 1 & 0 & : & 0 \end{bmatrix}$$

$$R_4 \leftrightarrow R_1$$

$$\begin{bmatrix} 0 & 1 & 0 & : & 0 \\ 1 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 3 & 4 & 3 & : & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 4 & 3 & 0 \end{array} \right]$$

$$R_4 \rightarrow R_4 - 3R_1 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 4 & 3 & 0 \end{array} \right]$$

$$R_4 \rightarrow R_4 - 4R_2 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 \end{array} \right]$$

$$R_4 \rightarrow R_4 - 3R_3 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$k_1 = 0, k_2 = 0, k_3 = 0$$

$\therefore$ The system has trivial solution, so the system shows that given vectors are linearly independent.

For What values of λ is the set $\{t+3, 2t+\lambda^2+2\}$ linearly independent
(PP-2016)

$$V = a_1 V_1 + a_2 V_2 + a_3 V_3 + \dots$$

$$(0, 0, 0) = a_1(t+3) + a_2(2t+\lambda^2+2)$$

$$(0, 0, 0) = (a_1 t + a_1 3) + (2a_2 t + a_2 \lambda^2 + 2a_2)$$

$$a_1 t + 2a_2 t = 0$$

$$a_1 t + 2a_2 t = 0 \quad \text{--- (1)}$$

$$3a_1 + a_2 \lambda^2 = 0 \quad \text{--- (2)}$$

$$2a_2 = 0 \quad \text{--- (3)}$$

from (3)

$$\boxed{a_2 = 0}$$

Put in equation (1)

$$\boxed{a_1 = 0}$$

~~(Put values in equation (2))~~

~~(2) is~~

$$\text{OR } 3a_1 + a_2 \lambda^2 = 0$$

$$3a_1 = -a_2 \lambda^2$$

$$\lambda^2 = \frac{3a_1}{-a_2}$$

$$\boxed{\lambda^2 = 0}$$